

Decoding DVZ the VZ300 DOS Floppy Disk Image Format
As used by DSVZ200 the VZ200 and VZ300 emulator

The VZ300 Double Density 5.25 Floppy Disk

Each disk has a 78K capacity, a total of 640 sectors per disk arranged as 40 tracks of 16 sectors per track in an interleaved arrangement. The image file size is 98,560 bytes. Track 0 (Sector 0 - 14) contains 112 directory entries; sector 15 of track 0 is a map of used sectors or write-protected sectors, leaving 624 free sectors (78K) for programs.

Image File Layout with interleaved sectors

	Track0	Track1	Track2	Track3	...
Sector0	\$9Abytes	\$9Abytes	\$9Abytes	\$9Abytes	...
Sector11	\$9Abytes	\$9Abytes	\$9Abytes	\$9Abytes	...
Sector6	\$9Abytes	\$9Abytes	\$9Abytes	...	
Sector1	\$9Abytes	\$9Abytes	...		
...	...	...			

Tracks

There are a total of 40 tracks per disk, each track is 2464 bytes with the image starting at track 0 sector 0, tracks are arranged 0 to 39 with 16 interleaved Sectors per Track.

Sectors

There are a total of 640 sectors per disk, sectors are 154 bytes arranged as in this Sector Order table.

Sector Order Table

0	11	6	1	12	7	2	13	8	3	14	9	4	15	10	5
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Sectors consist of four sections; a newly formatted image has only headers written on the image.

Sector Format

OFFSET	BYTES	SECTION
0	24	HEADER
24	126	DATA BYTES
150	2	NEXT TRACK AND SECTOR
152	2	16 BIT ADD CHECKSUM

Sectors begin with the sector header of 24 bytes, in each sector header there are three variables TR, SC and T+S. TR is the track, SC is the sector and T+S is track + sector used as a checksum.

Sector Header Format

GAP1	80	80	80	80	80	80	00				
IDAM	FE	E7	18	C3	TR	SC	T+S				
GAP2	80	80	80	80	80	00	C3	18	E7	FE	

Sector Checksum Algorithm

The data section of each sector has a checksum found at the last two bytes of each sector. The checksum is a simple 16bit add equation of the data section (excluding the sector header).

Sectors use the 16 bit add checksum method

(Checksum ++Byte) && \$FFFF

Sector Map

Track 0 sector 15 is the sector map, only the first 78 bytes are used the DOS ROM does not in any way use the remaining 50 bytes. The map sector should be found located at an offset 2310 bytes from the start of the image file, two bytes are used per track, one bit per sector; 0 empty and 1 used. The sector map begins at the first byte of the data section; bytes 24 to 102.

Note that the sector map does not include track 0 and the sector maps bits are not arranged out of order.

Map word

Low byte								High byte							
B7	B6	B5	B4	B3	B2	B1	B0	B7	B6	B5	B4	B3	B2	B1	B0
S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	SA	SB	SC	SD	SE	SF

Directory

Track 0 sector 0 to 14 holds the directory information for the disk. There are 112 entry places; 16 bytes per entry, 8 entries per sector.

Entry elements and their offsets

	filetype	delimiter	filename	track	sector	startaddr	filesize
bytes	1	1	8	1	1	2	2
offset	0	1	2	10	11	12	13

